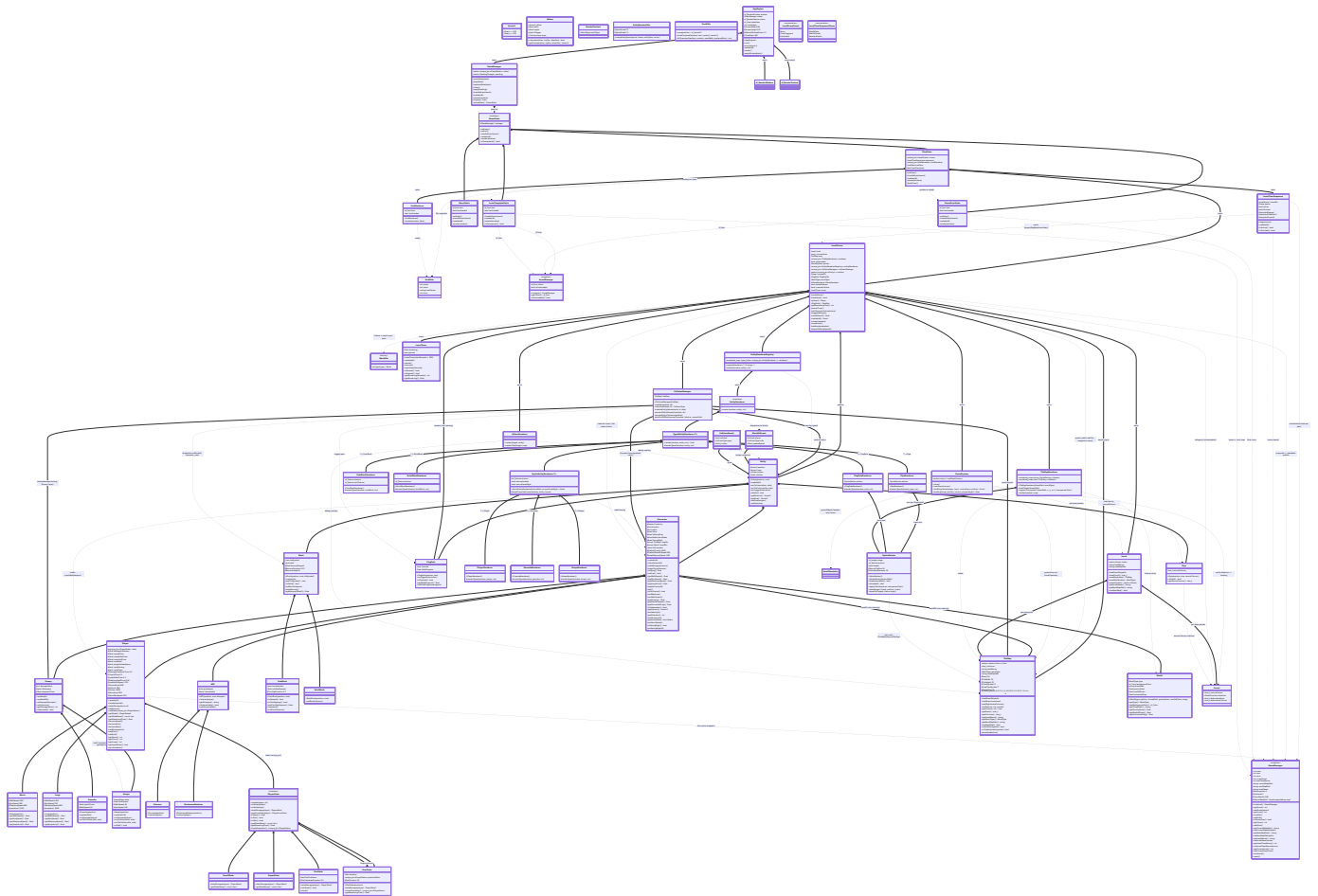


Full Project Class Diagram

Mermaid diagram covering all model, view, and controller classes (k.2.5). All three restructuring stages are reflected. Stage 1: model interface segregation. Stage 2: the goal castle is painted into the TileMap completion zone from a 21-symbol sheet (`TileMap::CastleSymbols` , registered by `TileMapRenderer` from atlas rows `y = 696/712` (tower) and `728/744/760` (base)); pipes spawn from 'p' / 'P' grid runs; tile collision treats castle symbols as ground/solid; and every static-image draw goes through the shared `SpritePainter` facade (the `Castle` entity and `CastleRenderer` are gone). Stage 3: the controller god-object split — the live level is `LevelScene` (update reports `Event s`), the flagpole clear play is `LevelClearSequence` , and `PlayState` is slimmed to state transitions, HUD snapshot and debug keys. Stage 3.1: the warp-pipe rules moved out of `LevelScene` into `PortalSystem` (entry detection, one-way inert columns, re-emergence placement), sharing the header-only `controller::geometry helpers ( isGroundSymbol / groundTopAt in LevelGeometry.h )` that `LevelScene` still uses for the castle paint. Companion doc: `CLASS_ENCAPSULATION_AND_CONNECTIONS.md` (per-class encapsulation + inter-class methods).



Enums

Enum	Values
CollisionType	None, Top, Bottom, Left, Right
CollisionLayer	Player, Enemy, Environment, Trigger
AnimState	Idle, Walk, Run, Jump, Fall, Die
PlayerAnimState	SmallIdle, SmallWalk, SmallRun, SmallJump, SmallFall, SmallDie, BigIdle, BigWalk, BigRun, BigJump, BigFall, FireIdle, FireWalk, FireRun, FireJump, FireFall, FireShoot, StarIdle, StarWalk, StarRun, StarJump, StarFall
KoopaState	Walking, ShellIdle, ShellSpinning
PortalDirection	Down, Up
WorldType	Overworld, Underwater, Castle

LevelScene::Event	None , ClearTriggered , RunEnded
LevelClearSequence::Phase	SlideToPole , WalkToCastle , ReachedCastle

Legend

Arrow	Meaning
< --	Inheritance
* --	Composition (owning)
o --	Aggregation (non-owning reference)
..>	Dependency / usage